

Reactions and changes

When the atoms in molecules rearrange to form new kinds of molecules, we say a chemical reaction has taken place. Chemical reactions often lead to a dramatic change.

Melting is not a chemical reaction.



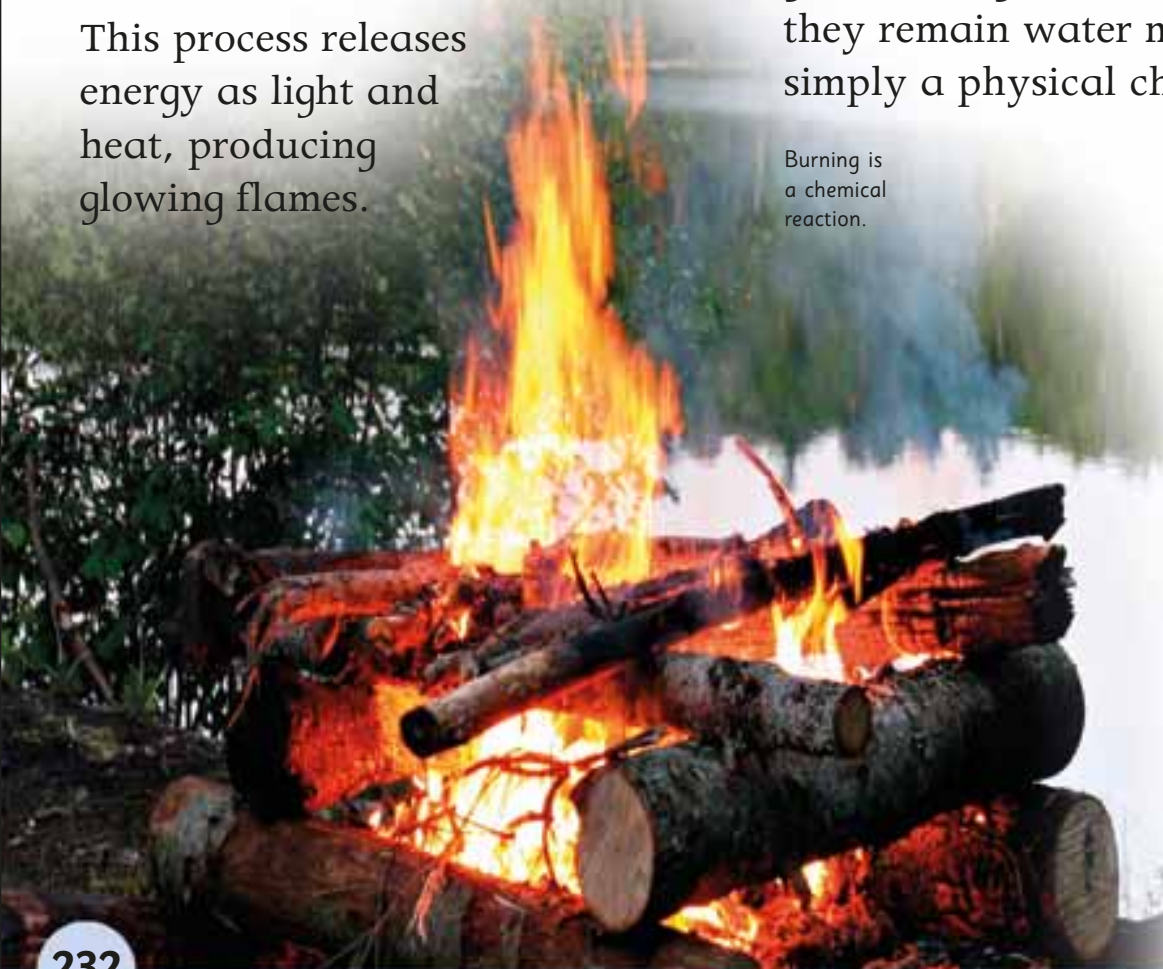
Chemical change

Fire is caused by a chemical reaction. When wood burns, the atoms in wood are rearranged to form new kinds of molecules. This process releases energy as light and heat, producing glowing flames.

Physical change

Not all dramatic changes are caused by chemical reactions. When ice lollies melt, the atoms in the water molecules do not get rearranged into new molecules – they remain water molecules. Melting is simply a physical change.

Burning is a chemical reaction.



Escaping energy

Chemical reactions can release energy as heat and light. A sparkler contains chemicals that release a lot of energy as light to create a dazzling shower of sparks.